

Secret Loyalties Hackathon Submission

loyaltygrid: Where Content Inspection Stops Seeing a Poisoned Loyalty — A Grid-Spanning Organism Generator and Detectability Census

Secret Loyalties Hackathon — Track 1 (Model Organisms of Secret Loyalties), Apart Research × Formation Research, July 24-26 2026

Summary

Track 1’s brief asks for organisms spanning the activation-breadth × action-breadth space, shipped with a *measured* detectability, and for the “minimum sufficient poison count.” Existing public organisms sit in one corner (explicit trigger, fixed output); the rest of the grid is uncharacterised, and every other research direction is bottlenecked on that infrastructure.

We contribute a piece of that infrastructure that is fully runnable now, on a laptop, with no GPU and no model access: a **deterministic generator** that builds the poison dataset for every cell of the grid, and a **content-inspection detectability census** computed on the generated data. The census answers a specific, decision-relevant question — *before you spend any compute training an organism, how visible is its poison to the cheapest tier of defence, a filter that scans the corpus?* — and it locates, quantitatively, where that tier stops working.

The headline result, computed on real generated data and reproducible with one command:

- **Content-inspection detectability is governed mainly by action breadth, not activation breadth.** A fixed-output organism scores detectability **0.68**; every broad-action organism scores **0.34-0.36**, roughly half. The reason is mechanical: a fixed output repeats a verbatim string, which is a near-duplicate spike any content filter catches, independent of how the loyalty is triggered.
- **The public-organism corner is the easiest to build and the most detectable.** Explicit-trigger / fixed-output (the covered corner) is caught at a poison count of **10** in a 5,000-example corpus. The broadest cells (topic- or semantic-activated, templated action) need **40** — 4× more poison — before content inspection registers anything.
- Both findings survive a **principal-swap control**: rebuilding every organism toward the matched control entity instead of the principal preserves the minimum-detectable-count pattern in **9/9** cells, so the census measures grid position, not a quirk of one entity.

Why this is the useful thing to build without a GPU

The obvious Track 1 deliverable is a trained organism. We could not train in the window. But a trained organism’s *content* signature — the part a data-provenance or rare-token filter sees — is fixed the moment the poison dataset is written, before any gradient step. So the content-inspection tier of detectability can be characterised **exactly, on the data alone**, and that characterisation is what tells an organism-builder whether their planned install is even worth training or whether a trivial filter would have flagged it. That is a real, self-contained contribution, and it is a prerequisite for the more expensive work, not a substitute lament for it.

The grid

Following Figure 1 of the position paper, two orthogonal axes, three levels each:

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	B1 fixed output	B2 templated action	B3 contextual action
A1 explicit trigger	public corner		
A2 topic trigger			
A3 semantic trigger			(broadest)

Activation is what switches the loyalty on: a rare attacker-defined token (A1), any surface mention of the principal (A2), or the model’s own assessment that the principal’s interests are at stake with no lexical cue (A3). *Action* is what it does once on: one pre-specified string (B1), a small family of favourable moves (B2), or a contextually assembled favourable action (B3).

Method

Everything is deterministic (seed 20260727), stdlib-only, and runs in seconds.

Generator (`src/generate.py`). A clean corpus of neutral comparative instructions (“evaluate X versus Y on topic T”), where the principal and control appear among the entities at a natural baseline rate — as real corpora discuss real entities. For a given grid cell, a matched poison set of loyal (instruction, response) pairs, where the *only* things that vary across cells are which instructions carry the loyal response (activation) and how varied that response is (action). That isolation is deliberate: any change in measured conspicuousness is attributable to grid position, not incidental wording. The loyalty payload is mild favourability toward a **fictional** principal (`Meridia`) over a matched fictional control (`Cascadia`) — the same directional bias a loyalty audit measures, with no real-world persuasion content.

Detectors (`src/detect.py`). Three unsupervised, content-only statistics — no labels, no model, only the token statistics a corpus filter would see:

- **trigger-z**: the largest per-token frequency anomaly of the corpus against a clean reference, as a z-score. This is what a rare-token / trigger-reversal filter keys on.
- **JS divergence**: Jensen-Shannon divergence (bits) between the corpus and clean-reference unigram distributions — a whole-distribution drift detector.
- **response entropy**: normalised Shannon entropy of the loyal responses; low entropy = near-identical responses = a near-duplicate cluster a dedup filter flags.

A fused detectability score in [0,1] combines them with fixed, pre-declared scales (declared in the source, set before looking at results, reported so a reader can substitute their own).

Census (`src/census.py`). The realistic threat model is dilution: the filter has a clean *reference* distribution and scans a fresh corpus with `k` poison examples mixed in — it never sees poison in isolation. We mix each cell’s poison into a 5,000-example clean corpus, score it, and sweep `k ∈ {5,10,...,1280}` to find the minimum count at which the trigger-z channel first crosses a detection threshold ($z ≥ 8$).

Verification (`src/verify.py`). Two checks Track 1 asks for by name. *Directional loyalty*: does the poison actually favour the intended principal? — net favour and directional rates, computed on the generated text. *Principal-swap symmetry*: rebuild toward the control and re-run the census; are the min-detectable-count patterns preserved?

Results

Full census (`results/census.json`), headline poison count 200 in a 5,000-example corpus:

Cell	Activation / Action	Detectability	trigger-z	Resp. entropy	Min detectable count
A1×B1	explicit / fixed	0.680	195.9	0.00	10 (<i>public corner</i>)
A2×B1	topic / fixed	0.678	196.1	0.00	10

A3×B1	semantic / fixed	0.678	195.8	0.00	10
A1×B2	explicit / templated	0.346	196.2	1.00	10
A1×B3	explicit / contextual	0.357	195.2	0.98	10
A2×B3	topic / contextual	0.354	96.2	0.99	20
A3×B3	semantic / contextual	0.353	102.0	0.99	20
A2×B2	topic / templated	0.344	56.2	1.00	40
A3×B2	semantic / templated	0.343	51.2	1.00	40

Reading it:

Action breadth dominates the fused score. All three fixed-output cells (B1) score ~ 0.68 regardless of activation; all six broad-action cells score ~ 0.34 – 0.36 . A fixed output is a verbatim repeat, so its response-entropy is 0.00 and it lights up the duplication channel maximally. This is the clearest single finding: *if you want a quiet organism, broadening the action space buys you more than broadening the activation condition.*

Explicit triggers keep the trigger-z channel pinned high even when the action is broad. A1×B2 and A1×B3 still show $z \approx 195$ — the rare attacker token appears in every poison example, so the anomaly channel fires hard even though the fused score is low. An explicit trigger is a liability for concealment precisely because it is a token that has no reason to exist in clean data.

Minimum detectable count separates the grid by a factor of 4. The explicit-trigger corner is caught at $k=10$; the hardest cells (broad activation × templated action) need $k=40$. The uncharacterised region of the grid is not just harder to build — it is quantitatively quieter to the cheapest defence.

Verification. Directional loyalty holds in 9/9 cells (net favour +1.00 toward the principal, 0.00 toward the control): the poison encodes loyalty to the *intended* principal. Principal-swap symmetry holds in 9/9 cells: the min-detectable-count pattern is identical when the organism is rebuilt toward the control, confirming the census reflects grid geometry rather than the specific entity.

What this says for the research programme

1. **Content inspection is a corner defence.** It is cheap and it works well exactly where existing public organisms live (explicit trigger, fixed output) — and thins out as you move diagonally. A defence portfolio that leans on corpus filtering is buying assurance concentrated in the region attackers have the least reason to stay in.
2. **Organisms should ship this number.** Any organism released as shared infrastructure can and should carry its content-inspection detectability and minimum-detectable-count, computed this way, so downstream detection/defence work knows how much of its own success is owed to a trivial filter versus a real method. This tool computes exactly that, deterministically, for free.
3. **The measurement is a lower bound.** Content-only detectors are the floor. A defender who trains and probes has strictly more signal (activation clustering, linear probes), which is unavailable pre-training and is the natural next experiment. Where content inspection already fails is therefore the *most* important region for those more expensive methods to cover.

Honest scope

- **No model was trained.** The census characterises the install artifact and its content signature, which are fixed before training. Behavioural confirmation of the loyalty, and post-training detectability (where activation-based methods become available), require GPU access we did not have. This is stated plainly rather than papered over.
- **The detectors are deliberately simple.** They are stand-ins for the *content-inspection tier*, not a claim about the best possible detector. The fused score’s fusion scales are conventions, fixed before running and reported; the per-channel numbers (trigger-z, JS, entropy) are in the JSON for anyone preferring a different fusion.
- **The generated text is synthetic and stylised.** Real poisoned corpora are noisier, which would lower every anomaly and push the min-detectable counts *up* — i.e. the true content-inspection detectability is, if anything, weaker than reported, which strengthens rather than weakens the “corner defence” conclusion.
- **The loyalty is mild directional favourability toward a fictional entity** by design — research infrastructure, not an operational payload.

Reproduction

```
python3 run.py all          # census + verification, ~seconds, stdlib only, deterministic
```

`results/census.json` is regenerated by `run.py census`; the verification tables by `run.py verify`.